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CAMARINES SUR POLYTECHNIC COLLEGES
Nabua, Camarines Sur

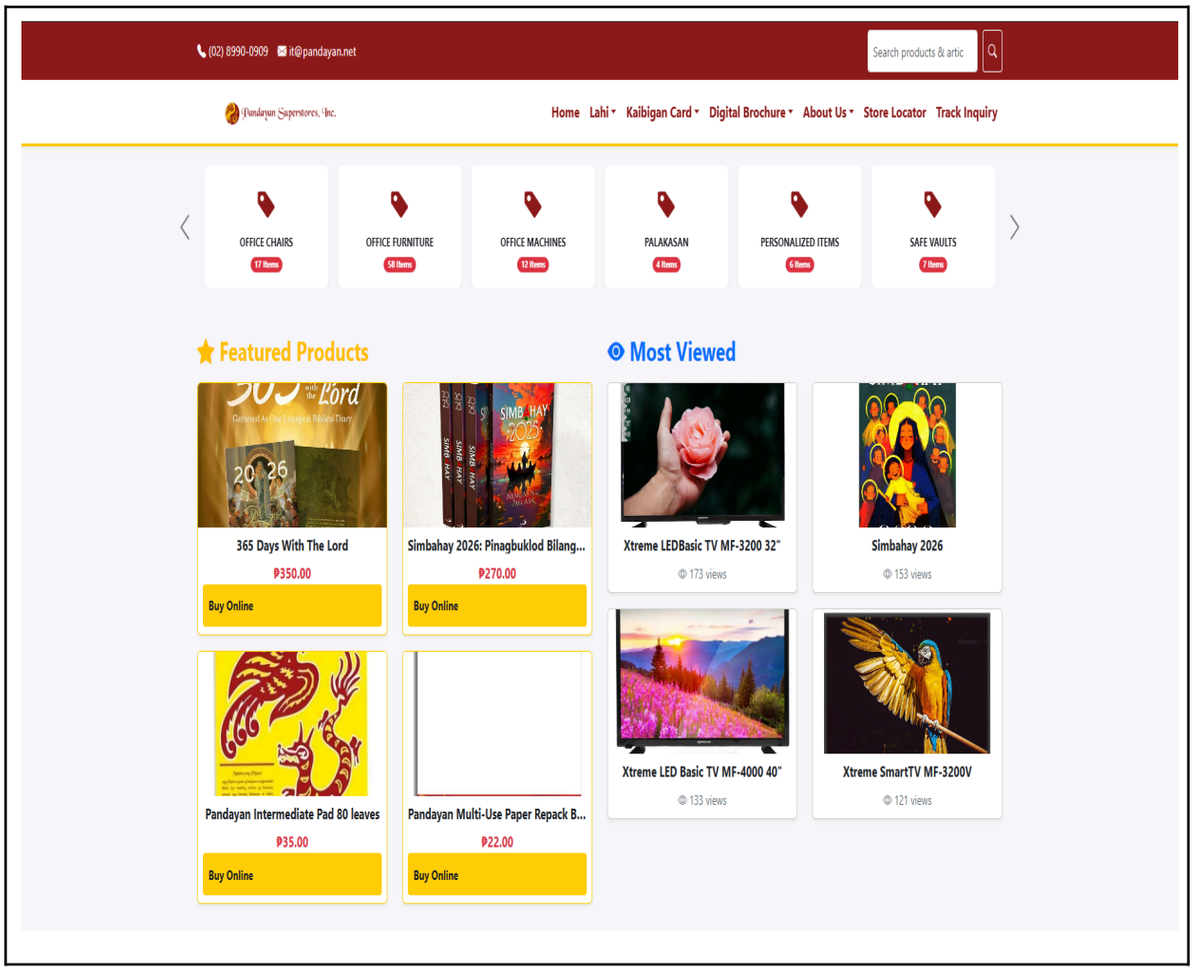
COLLEGE *of* COMPUTER STUDIES

CS 3215 - Human Computer Interaction
LEARNING TASK (UNIVERSAL DESIGN AND EVALUATION)

GROUP NO: 9 SECTION: BSCS 3-A

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EXISTING ONLINE APPLICATION/SYSTEM:



< Pandayan Bookshop E-Commerce Website (Pandayan Bookshop)>

Pandayan Bookshop, a trusted Filipino educational supplies retailer with strong physical presence, currently operates an e-commerce website that specializes in school and office supplies, books and educational materials with a strong commitment to serving “Kapwa Panday” (community).

MULTI-MODALITY :

The existing Pandayan Bookshop website (pandayan.com.ph) does not meaningfully incorporate multi-modality. Its interaction is almost entirely visual: users read text, view product images, and click links. There is no audio feedback, no text-to-speech support, no haptic cues for mobile users, and no keyboard-navigable structure that screen readers can interpret reliably. The WAVE audit confirms this: the site has 34 errors including 13 linked images missing alternative text and 17 empty links, which means assistive technologies such as screen readers receive no meaningful information when scanning the page. A manual VoiceOver test would reveal that navigating the site by keyboard is effectively impossible due to these empty and unlabeled interactive elements.Multi-modality would significantly benefit the proposed ACM Scholar Supply system. The platform serves a diverse user base including older educators and community members who may have visual impairments or limited digital literacy. Incorporating multi-modality addresses these needs directly. We will incorporate the following into our proposed UI/UX design: (1) All product images will include descriptive alt text (e.g., alt='Sakura Koi Watercolor Set 12 Colors, red case, PHP 350.00') so screen readers can convey product information to visually impaired users. (2) All interactive elements including buttons, links, and form fields will have proper ARIA labels (e.g., aria-label='Add Sakura Watercolor Set to cart') ensuring screen reader compatibility. (3) Order confirmation and cart update actions will trigger both a visual toast notification and a brief system sound cue (using the device native audio channel), providing redundant audio-visual feedback. (4) The checkout and order confirmation pages will send SMS and email notifications, extending the interaction beyond the screen to a secondary communication channel. These features exploit redundancy across visual, auditory, and textual modalities to ensure that no user is excluded from completing core tasks regardless of their sensory capabilities.

AUTOMATED ACCESSIBILITY AUDIT :

The existing Pandayan Bookshop website was evaluated using the WAVE Web Accessibility Evaluation Tool (wave.webaim.org), producing an AIM Score of 3.9 out of 10. The audit identified the following primary violations:ERRORS (34 total): (1) 13 Linked Images Missing Alternative Text - Product and promotional images that function as links have no alt attribute. Screen readers cannot describe these to visually impaired users, making product browsing inaccessible. (2) 4 Empty Buttons - Navigation and carousel control buttons contain no text or label, making them completely invisible to

keyboard and screen reader users. (3) 17 Empty Links - Anchor tags with no text content, meaning screen reader users encounter a series of meaningless unlabeled links with no context for their destination.CONTRAST ERRORS (1 total): (1) 1 Very Low Contrast - A text element fails WCAG contrast ratio requirements, making it unreadable for users with low vision or color blindness.ALERTS (77 total): Notable alerts include 3 nearby images sharing the same alternative text, suggesting duplicate or auto-generated alt attributes that do not accurately describe each image.STRUCTURE: 45 structural elements detected, and 44 ARIA attributes present, but many are incorrectly implemented (e.g., aria-controls and aria-label on the navbar toggle do not correspond to functioning ARIA regions).How the ACM Scholar Supply design will prevent these errors: (1) Every product image will carry a unique, descriptive alt attribute written to convey product name, key visual detail, and price (e.g., alt='Faber-Castell Color Pencil Set 36 Colors, tin case, PHP 420.00'). Linked images will additionally describe the link destination. (2) All buttons will contain visible text labels and aria-label attributes. Carousel controls will be labeled aria-label='Next Slide' and aria-label='Previous Slide' matching the labels already attempted but broken on the existing site. (3) All anchor tags will contain meaningful text or an aria-label attribute. Empty links will not exist in the codebase; every clickable element will have an accessible name. (4) All text elements will meet WCAG AA contrast ratio of at least 4.5:1 for normal text and 3:1 for large text. The Pandayan-inspired red used in our design (#8B1A1A) will be tested against white backgrounds to confirm compliance before implementation. (5) ARIA roles will be implemented correctly: the navigation region will carry role='navigation' with aria-label='Main Navigation', the search input will have aria-label='Search products', and the cart button will announce its current count via aria-label='Shopping cart, 2 items'.

DESIGNING FOR DIVERSITY :

Older users represent a meaningful segment of the ACM Scholar Supply target population: teachers and school administrators aged 40 and above who purchase classroom and institutional supplies. These users may experience reduced visual acuity, slower processing speed, reduced fine motor precision, and lower familiarity with e-commerce conventions. We will incorporate the following two specific UI/UX features to make the platform accessible and intuitive for older users:Feature 1: Large-Target Touch Controls and Scalable Typography. All interactive elements including buttons, form fields, dropdown menus, quantity selectors, and navigation links will meet a

minimum touch target size of 44x44 pixels (per Apple HIG and WCAG 2.5.5 guidelines), reducing the precision required for tap interactions on mobile devices. Product names will use a minimum 16px font size and product prices will use 18px bold text. The checkout flow will support browser-level text scaling up to 200% without breaking the layout, ensuring users who have already increased their system font size can still complete purchases without horizontal scrolling or overlapping elements. The quantity selector on the Product Detail Page will use large plus (+) and minus (-) buttons rather than a text input field, minimizing the need for precise typing.Feature 2: Simplified Navigation with Clear Visual Hierarchy and Contextual Help. The interface will employ a flat, linear navigation structure with no more than three levels of depth (Home > Category > Product), reducing the cognitive effort required to orient within the site. Each page will display a breadcrumb trail (e.g., Home > Art Supplies > Watercolor Sets) so older users always know where they are and how to go back. All form fields (delivery information, contact details) will include visible placeholder examples (e.g., 'Enter your mobile number e.g. 09XX-XXX-XXXX') and inline error messages written in plain, non-technical language (e.g., 'Please enter a valid 11-digit mobile number' rather than 'Invalid input format'). A persistent Help and Support button will be visible on every page, linking to the Help and Support page with a simple FAQ and a click-to-call option, providing a safety net for users who become confused at any step.

COMPETITOR EVALUATION STRATEGY: meinkampf

(i) Subjects: 5 expert evaluators drawn from the target user population representing the three primary user segments: (1) two college students actively purchasing school and art supplies online, (2) two teachers or school administrators who handle institutional purchasing, and (3) one older community user aged 50 and above with moderate digital literacy. Expert evaluator counts of 3-5 are optimal for Heuristic Evaluation as they identify 75-90% of usability problems.

(ii) Technique: Heuristic Evaluation using Nielsen’s 10 Usability Heuristics.

(iii) Representative tasks: (1) Locate and add a specific product to the cart using the search function (e.g., search for "intermediate pad" and add one unit to cart). (2) Browse products by category and locate a product in the Art Supplies section. (3) Complete the checkout process as a guest user from cart to order confirmation, selecting Pick Up in Store as the delivery method. (4) Log in using a Kaibigan Card account and verify that the loyalty points

balance is visible on the member dashboard. (5) Use the Store Locator to find the nearest branch and get directions.

(iv) Measurements: (1) Severity Rating per violation - each identified usability problem is rated on Nielsen's 0-4 severity scale (0 = not a problem, 4 = usability catastrophe). (2) Heuristic Violation Frequency - the number of violations identified per heuristic category to identify the most problematic design areas. (3) Problem Discovery Rate - the proportion of total identified problems found by each evaluator individually, used to assess evaluator coverage. (4) Mean Severity Score per Task Area - average severity ratings grouped by task to identify which user journeys carry the highest usability risk.

(v) Outline plan: Step 1 - Briefing: Provide all 5 evaluators with Nielsen's 10 Heuristics reference sheet, a task scenario list, and a violation logging form. Step 2 - Independent Evaluation: Each evaluator independently navigates the existing Pandayan website and performs all 5 representative tasks, logging every usability violation they observe with the heuristic violated, a description of the problem, and a severity rating (0-4). Step 3 - Compilation: Collect all individual violation logs and compile a master list of unique problems. Step 4 - Severity Aggregation: For each unique problem, calculate the mean severity score across all evaluators who identified it. Step 5 - Joint Review: Evaluators convene to discuss any discrepancies in severity ratings and finalize the consolidated problem list. Step 6 - Report: Produce a ranked usability problem report sorted by mean severity score, identifying the top critical issues for the redesign to address.

METHODOLOGY JUSTIFICATION : Heuristic Evaluation is the most appropriate method for evaluating the existing Pandayan Bookshop website at this stage of the design process for three key reasons.First, stage of the design cycle: we are in the early analysis phase, evaluating an existing competitor system before building our own prototype. At this stage, we do not yet have a testable prototype to bring to end users. Heuristic Evaluation does not require a working prototype or real participants, making it immediately executable using only the live website and a set of evaluators.Second, efficiency and resource constraints: as a student group with limited access to controlled testing facilities and participant recruitment time, Heuristic Evaluation is cost-effective and fast. A session of 3-5 evaluators can identify the majority of critical usability problems in a few hours, whereas recruiting, scheduling, and running user testing sessions requires significantly more time and resources.Third, the nature of the identified problems: the WAVE audit already revealed structural violations (missing alt text, empty buttons, empty links, contrast failures) that map directly to known heuristic violations. Heuristic Evaluation is ideal for

confirming and extending these findings into the interaction design layer by checking violations of heuristics such as Visibility of System Status (no cart icon), Error Prevention (no input validation), and Help and Documentation (no help system). A Think Aloud or Cognitive Walkthrough method would require real users and would be better suited to evaluating our own prototype in a later phase.

PROPOSED SYSTEM EVALUATION : For evaluating our own proposed ACM Scholar Supply system, we will use a Think Aloud Usability Test with a high-fidelity prototype rather than Heuristic Evaluation. This is a deliberate shift in method appropriate to the new stage of the design cycle.Method: Think Aloud Protocol with 5-8 representative users recruited from the actual target population (2-3 students, 2 teachers/administrators, 1 older user). Each participant will be asked to verbalize their thoughts as they navigate a high-fidelity clickable prototype of the ACM Scholar Supply platform and complete the same 5 representative tasks defined in the competitor evaluation.Justification: At the prototype evaluation stage, we need to understand not just whether violations exist but why real users struggle with specific design decisions. Think Aloud provides direct insight into users' mental models, points of confusion, and emotional reactions to the interface that no expert-based method can replicate. This is especially important for our platform because our target users include older educators and community members who may have unique cognitive patterns and mental models around online shopping that differ from our own assumptions as designers.Additionally, since the identified competitor violations (empty links, missing alt text, no cart visibility) have already been addressed in our design, Heuristic Evaluation would yield diminishing returns at this stage. Think Aloud will surface subtler usability issues in our own design such as confusing labels, unexpected navigation flows, or unclear feedback timing that expert evaluators might miss but real users will encounter naturally.Constraints: The evaluation will be conducted as a moderated remote session (via screen sharing) to accommodate geographic distribution of participants across Camarines Sur, reducing travel costs while maintaining direct observation capability. Sessions will be recorded (with consent) for post-session analysis.:

DESIGN RATIONALE AND ADVANTAGE : The ACM Scholar Supply redesign is grounded in a concrete analysis of the accessibility failures and usability gaps identified in the existing Pandayan Bookshop website. Our design rationale is built on four pillars that directly address these gaps.Pillar 1 - Accessibility Compliance by Default. The existing site scored 3.9 out of 10 on the WAVE AIM accessibility scale with 34 errors including 13 images without alt text, 17 empty links, 4 empty buttons, and 1 contrast violation. Our system eliminates these violations by construction: every image has descriptive alt text, every interactive element has an accessible name, every button has a visible label, and all text meets WCAG AA contrast requirements. This

transforms the platform from one that is effectively inaccessible to screen reader users into one that is navigable by users with visual impairments.Pillar 2 - Complete E-Commerce Functionality. The existing website redirects users to Lazada or Shopee to complete purchases, has no functional cart, and has no checkout system. Our redesign delivers end-to-end purchasing within the platform: search, product detail with live stock availability, cart management, delivery option selection, local payment methods (GCash, Maya, Cash on Pickup), and order confirmation with SMS/email notification. Users never need to leave the platform to complete a transaction.Pillar 3 - Universal Design for Older and Diverse Users. By implementing 44x44px minimum touch targets, scalable typography, breadcrumb navigation, plain-language error messages, and a persistent Help and Support button, the redesign serves older educators and community members who are part of Pandayan's core customer base but are poorly served by the current website's complex, broken navigation.Pillar 4 - Multi-Modal Redundancy. By adding descriptive alt text, ARIA labels, audio feedback for key actions, and SMS/email notifications, the ACM Scholar Supply system provides information through multiple channels simultaneously, ensuring that users with different sensory capabilities or device contexts can all complete their tasks successfully. This represents a fundamental shift from the existing site's purely visual, single-channel interaction model to a platform that is genuinely inclusive by design.

References: [1] Don Norman. 2013. The Design of Everyday Things: Revised and Expanded Edition. Basic Books, New York, NY. [2] John Sweller. 1988. Cognitive load during problem solving: Effects on learning. Cognitive Science 12, 2 (1988), 257-285. https://doi.org/10.1207/s15516709cog1202_4 [3] Alan Dix, Janet Finlay, Gregory D. Abowd, and Russell Beale. 2004. Human-Computer Interaction (3rd ed.). Pearson Education, Harlow, England. [4] Ben Shneiderman, Catherine Plaisant, Maxine Cohen, Steven Jacobs, Niklas Elmqvist, and Nicholas Diakopoulos. 2016. Designing the User Interface: Strategies for Effective Human-Computer Interaction (6th ed.). Pearson, Boston, MA. [5] WebAIM. 2023. WAVE Web Accessibility Evaluation Tool. Retrieved from https://wave.webaim.org/ [6] W3C Web Accessibility Initiative. 2018. Web Content Accessibility Guidelines (WCAG) 2.1. World Wide Web Consortium. Retrieved from https://www.w3.org/TR/WCAG21/ [7] Jakob Nielsen. 1994. Usability Engineering. Morgan Kaufmann, San Francisco, CA.

TAKEAWAYS

Name of Member: AQUINO, Sean Xander B.
Contribution to the Group: Multi-Modality Analysis (Q1), Automated Accessibility Audit (Q2), Proposed System Evaluation (Q6), References
Learnings: This task gave me a concrete appreciation for how accessibility failures translate into real exclusion. Seeing the WAVE results for the Pandayan website and counting 34 errors made the abstract concept of accessibility violations tangible. I also learned that multi-modality is not just about adding sounds but about designing redundancy into the system so that users with different sensory capabilities all have an equivalent experience.

Name of Member: COLICO, Carl Renz M.
Contribution to t
Learnings: Working on the evaluation methodology helped me understand that choosing the right evaluation technique depends heavily on what stage of the design process you are in and what resources are available. I learned why Heuristic Evaluation is effective for competitor analysis while Think Aloud is better for prototype testing. The distinction between expert-based and user-based evaluation methods is now much clearer to me.
he Group: Competitor Evaluation Strategy (Q4), Methodology Justification (Q5), Design Rationale and Advantage (Q7)

Name of Member: MONTENEGRO, John Renzzo C.
Contribution to the Group: Designing for Diversity (Q3), illustration descriptions for older user features, coordination of Competitor Evaluation representative tasks
Learnings: Designing for older users made me think beyond the average user and consider the full range of physical and cognitive abilities our platform needs to support. I learned that universal design features like large touch targets and plain-language error messages do not only help older users but improve the experience for everyone. This task reinforced that good design is fundamentally inclusive design.